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ONTARIO
SUPERIOR COURT OF JUSTICE

B E T W E E N:	)	
DAVID C. CHAN, BETTY JUNG CHAN, STEPHEN C. CHAN and ANDREW DAVIS CHAN	)	
Plaintiffs	)	*Roger G. Oatley & Kathryn L. Knight*, for the Plaintiffs
- and -	)	
THE ERIN MILLS TOWN CENTRE CORPORATION, LOBLAWS COMPANIES LIMITED, LOBLAWS INC. and LOBLAWS SUPERMARKETS LIMITED	)	*S. Wayne Morris & Delora Deravi*, for the Loblaws Defendants
Defendants	)	
	)	**HEARD:** June 6, 7, 8, 9, 10, 13, 14, 15, 16, 17, 20, 21, 22, 24, 27, 2005

LAX J.

[1] The main controversy in this action is the relationship between a slip and fall accident and the onset of a progressive neuromuscular condition known as Post Polio Syndrome.

Overview

[2] The plaintiff, David Chan, is a polio survivor who was born in Canton, China in 1941. As a very young child, he contracted poliomyelitis that affected his lower limbs and left him with a

significant impairment in his right leg. He learned to walk again, but with a brace on his weak right leg and with the support of a cane. He came to the United States for his education and trained as a neuropsychologist. He then settled in Vancouver where he established a successful clinical and consulting practice, married and raised a family and enjoyed a busy and fulfilling personal and professional life.

[3] During a visit to Toronto with his family in 1993, he slipped and fell in a Loblaws store and fractured the patella of his functional left leg. After the accident, he began to experience symptoms of profound fatigue, pain and muscle weakness and was eventually diagnosed with Post Polio Syndrome or PPS. Dr. Chan now wears leg braces on both legs and either uses forearm crutches to ambulate or a scooter. His level of activity is diminished and is expected to decline further.

[4] It is the plaintiffs' position that Loblaws was negligent and that the trauma and its consequences either caused PPS or contributed in a material way to Dr. Chan's overall impairment. Loblaws denies that it was negligent. If negligence is found, Loblaws concedes that the patellar fracture resulted in degenerative changes to Dr. Chan's left knee, but denies that the accident caused Dr. Chan's present condition. Loblaws submits that Dr. Chan would have developed PPS regardless of the slip and fall and that opinion evidence of the relationship between the trauma and the onset of PPS is unreliable, unnecessary and therefore, inadmissible.

What is PPS?

[5] I will introduce the subject with some background information that is largely drawn from a book chapter authored by Dr. Lauro S. Halstead.[1] He is commonly regarded as a pioneer on the subject of PPS and an expert in this area.

[1] Lauro S. Halstead, M.D., M.P.H., "Diagnosing Postpolio Syndrome: Inclusion and Exclusion Criteria" in J.K. Silver & A.C. Gawne, eds., *Postpolio Syndrome* (Philadelphia: Hanley & Belfus, 2004) 1 [Ex. 78A] [L.S. Halstead, "Diagnosing Postpolio Syndrome"].

[6] Paralytic poliomyelitis is an illness characterized by high fever, followed by muscular paralysis, and is caused by infection with the poliovirus. Historically, polio was divided into three stages: acute illness, recovery and stable disability. Although the late effects of polio have been recognized for more than a century, it was not until the 1970s that reports began to appear in the medical literature that persons who had paralytic poliomyelitis many years earlier were experiencing new health problems relating to their prior illness. By the mid-1980s, there was growing recognition among clinicians and researchers that polio had a fourth stage. Although this has been described by various terms, including “pospolio sequelae”, Dr. Halstead and most other medical researchers and clinicians use the term “postpolio syndrome” or PPS.

[7] Post Polio Syndrome first reached medical consciousness when the polio survivors of the epidemics in the 1940s and 1950s began to report new symptoms in the late 1970s, 1980s and 1990s, following a long period of stable functioning. According to Dr. Halstead, it soon became clear that there was a smaller group of specific and discrete symptoms that “most experts agree share a common pathophysiology primarily involving a dysfunction of polio-affected motor units.”[2] Typically, these problems occur after a period of functional and neurologic stability of at least 15 years after the initial episode of polio and include new weakness, fatigue, decreased endurance, and loss of function.

[8] Different theories are advanced for the pathogenesis of PPS. One theory is that the recovery process from paralytic poliomyelitis is characterized by terminal axonal sprouting from surviving, but weakened motor neurons. It is believed that over time, the neurons are unable to support muscle fibres and degeneration of terminal axons occurs due to an increased demand on each motor neuron to innervate more muscle cells beyond the capacity of the neuron to function. Thus, decades after recovery from acute paralytic polio, some polio survivors begin to experience new symptoms. Once post-polio weakening begins, the usual course is progressive decline at the rate of between one and seven per cent a year.

[2] *Ibid*. at 1.

[9] Until recently, the late effects of polio remained an obscure and largely unexplored area of medicine. Most of the scientific research was devoted to early management and prevention and with the advent of vaccines, interest and funding in polio-related problems waned. The disease appeared to have been conquered when new neurological changes appeared in polio survivors thirty to forty years after the epidemics. By then, organizations such as the American March of Dimes had turned its attention away from polio.

[10] Post Polio Syndrome is not as well known or as well understood as other neuromuscular diseases. There continues to be controversy about its diagnosis, pathogenesis and treatment. Nonetheless, I am satisfied that within the medical community, and particularly among those who have clinical experience with PPS patients, it is generally accepted that PPS is a progressive neurological disorder that produces a cluster of recognized and disabling symptoms in individuals who had poliomyelitis many years earlier.

The Admissibility of the Expert Opinion Evidence

[11] Admission of expert opinion evidence depends upon the application of the four criteria described by Sopinka J. in *R. v. Mohan*, [1994] 2 S.C.R. 9. These criteria are: (1) relevance; (2) necessity in assisting the trier of fact; (3) absence of any exclusionary rule; and (4) a properly qualified expert. Early in the trial, Loblaws brought a motion to exclude the opinion evidence of Dr. Neil Cashman linking trauma and PPS on the basis that the opinion was "novel scientific theory" and did not meet the admissibility criteria of relevance and necessity. There was no practical or efficient way to hold a "Daubert Hearing": see *Daubert v. Merrell Dow Pharmaceuticals, Inc., 509 U.S. 579 (1993)*. I therefore decided that I would hear the evidence during the course of the trial and, at its conclusion, determine the admissibility of the evidence.

The Relationship Between Trauma and PPS

[12] The evidence that is challenged can be briefly summarized. Initially, Dr. Chan's treating physicians, Dr. House and Dr. van Rijn, were not convinced that Dr. Chan's complaints of fatigue and muscle weakness were due to PPS. Dr. Frank Lipson, a physiatrist, was the first to

provide a diagnosis when he assessed Dr. Chan at the request of the defendants in August 2002. However, he thought that Dr. Chan's PPS developed independently of the 1993 accident.

[13] The plaintiffs then retained Dr. Neil Cashman. He is a neurologist and former Director of the Postpolio Clinic at the Montreal Neurologic Institute and Hospital. He assessed Dr. Chan in February 2003 and delivered a report in November in which he wrote:

> Over the course of 20 years with studying and treating post polio patients, it is my experience that this syndrome can frequently be dated from a traumatic event, such as an accident or a limb fracture. *This has been also documented in unpublished research performed by myself and Dr. Daria Trojan of McGill University, my trainee in this area.* (italics added)

[14] Loblaws retained Dr. Richard Bruno to respond to Dr. Cashman's opinion. He is a psychologist and Director of the International Center for Post-Polio Education and Research at Englewood Hospital and Medical Center in New Jersey. He denied that there was any published article in the medical literature documenting that PPS can be triggered by a fall.

[15] During Dr. Cashman's cross-examination, he acknowledged that there is essentially little or no literature linking trauma and its consequences with the onset of PPS and that Dr. Trojan's thesis pointed to the need for further study. Nonetheless, in his experience, PPS can frequently be dated from a traumatic event such as an accident or limb fracture. Loblaws says that at its highest, this establishes a temporal relationship between the accident and the development of PPS symptoms. It submits that Dr. Cashman's opinion of causal connection rests on unproven and untested assumptions, falling into the category of "novel scientific theory".

[16] Dr. Halstead has written about the relationship between trauma and PPS:

> Most commonly, the onset of new problems is gradual, but in many persons, the outset may be triggered by specific events such as a minor accident, a fall, a period of bed rest or surgery. Typically, individuals report that a similar event

> several years earlier would not have caused the same decline in health and function.[3]

[17] Dr. Bruno initially denied the existence of published literature documenting the relationship between trauma and PPS, but during his cross-examination he was confronted with extracts from his book, *The Polio Paradox* (New York: Warner Books, 2003), from articles he has authored and from information published on his website in which he identifies recent trauma as a risk factor for the onset of PPS. He too has been engaged in research that attempts to document the relationship and conducted a review of the histories of 244 patients evaluated at his hospital who had no other conditions that might cause new fatigue, weakness or pain. Almost 20 per cent of those surveyed reported that their PPS began after a traumatic event. On his website, Dr. Bruno refers to this survey and to himself as an expert witness who can provide a medical/legal consultation and is available to testify that PPS is sometimes accident-related. Ultimately, Dr. Bruno acknowledged that trauma could severely stress polio-damaged motor neurons and lead to PPS. This is the same view expressed by Dr. Halstead who is acknowledged as one of the leading experts on PPS.

[18] Despite the paucity of scientific research establishing the relationship between trauma and PPS, Dr. Cashman is certainly not alone in advancing this theory. It appears that the scientific community has been aware of the relationship for some years. Dr. Halstead relies on a study conducted twenty years ago.[4] In *Samples v. Wellesley Hospital*, 1987 CarswellOnt 2326 (H.C.J.) (eC), aff'd [1989] O.J. No. 1644 (C.A.) (QL), the plaintiff, a polio survivor, developed PPS symptoms after a fall in which she fractured her arm. Carruthers J. accepted the expert medical evidence that it was the fracture and resulting immobilization that led to the onset of PPS. This case was decided eighteen years ago.

[19] The impugned evidence is not "junk science" as I understand this term. Nor am I persuaded that Dr. Cashman's opinion falls into the category of "novel scientific theory". It is

[3] *Ibid*. at 5.
[4] *Orthopaedics* 8:845-850: 1985.

more properly characterized as expert medical opinion of the kind that is commonly admitted in injury cases, as it was in *Samples*, and then assessed on the basis of weight. Having said this, Dr. Cashman acknowledged that there has been little research to document the relationship between trauma and PPS and it has not been subjected to scrutiny in peer-reviewed publications. For these reasons, it may suffer from some of the same frailties as “novel scientific theory”: see *Daubert* at 15-16 and *R. v. J.-L.J.[R. v. J.J.]*, [2000] 2 S.C.R. 600 at 615. I am therefore prepared to assume that the opinion evidence should be approached as “novel scientific theory” and subjected to the analysis that follows.

Expert Opinion Evidence

[20] The general rule with regard to expert evidence is that it is admissible to assist the fact-finder in drawing inferences in areas where the expert has relevant knowledge or experience beyond that of the lay person: *R. v. Lavallee*, [1990] 1 S.C.R. 852 at 889. Expert evidence is permitted as an exception to the usual rule excluding opinion evidence in recognition of the fact that the average person, even if given information, may not possess the necessary knowledge in some cases to assess its significance or draw the correct inferences in a particular context: *R. v. Abbey*, [1982] 2 S.C.R. 24 at 42. The expert’s function is to provide a “ready-made inference” where the judge and jury, unassisted, are unable to form their own conclusions.

[21] The dramatic growth of experts, particularly in the social and behavioural sciences in criminal cases, led to debate about the proper role of the expert and the proper role of the judge in admitting expert evidence. This set the stage for a consideration of these issues in *Mohan*.

Novel Scientific Evidence

[22] *Mohan* establishes that expert evidence which advances a novel scientific theory or technique is subjected to special scrutiny to determine whether it meets a basic threshold of reliability (although relevant, its value may not be “worth what it costs”) and whether it is essential (i.e. necessary) in the sense that the trier of fact will be unable to come to a satisfactory

conclusion without the assistance of an expert. The closer the evidence approaches an opinion on the ultimate issue, the stricter the application of this principle.

[23] Novel scientific evidence has been variously described but, in general, it is science producing evidence or bodies of knowledge that are new, untested, and that deviate from accepted standards or lack any standards. As scientific knowledge is constantly changing, a case-by-case evaluation is necessary: *R. v. J.J.* at 616. The judge acts as a gatekeeper and screens the evidence for threshold reliability before admitting it.

[24] The threshold test of reliability is a cost/benefits analysis of the impact of the evidence on the trial process and its potential to mislead the trier of fact, particularly a jury. As Sopinka J. explained in *Mohan* at 21, "Dressed up in scientific language which the jury does not easily understand and submitted through a witness of impressive antecedents, this evidence is apt to be accepted by the jury as virtually infallible and as having more weight than it deserves."

[25] In *R. v. Melaragni* (1992), 72 C.C.C. (3d) 348 (Ont. Ct. Gen. Div.), (quoted with approval in *Mohan*), Moldaver J. (as he then was), spoke of the potential to overwhelm the jury with the "mystic infallibility" of the evidence or to confuse and confound it.

[26] In *R. v. M.(B.)* (1998), 42 O.R. (3d) 1 at 25 (C.A.), Rosenberg J.A. sounded a cautionary note about the exclusion of relevant evidence, but stated that in weighing the evidence, the trial judge must pay particular attention to whether: (1) the evidence would involve an inordinate amount of time that is not commensurate with its value; and, (2) the evidence is misleading in the sense that its effect on the jury is out of proportion to its reliability.

[27] In *R. v. Terceira* (1998), 38 O.R. (3d) 175 (C.A.), aff'd [1999] 3 S.C.R. 866, Finlayson J.A. also sounded a cautionary note. He emphasized that in determining threshold reliability, the trial judge must not decide ultimate reliability, which is the function of the trier of fact. Rather, he or she must restrict the inquiry to determining whether the proposed novel scientific technique or theory has a foundation in science. The threshold is met when the trial judge having reviewed the evidence feels that it is sufficiently reliable to be put to the jury for its review. Finlayson J.A.

stated, at page 21, that the process of arriving at this point is a flexible one: " … it would be unwise to define the threshold test of reliability with the precision advanced by the scientific community. Rather, the threshold test of reliability must remain capable of adaptation to changing circumstances and realities."

[28] The principles established in *Mohan* and discussed in other judgments have general application, but they must be viewed in the context in which they arose. In *Mohan*, the Court accepted the conclusion of the trial judge that the evidence of a psychiatrist should be excluded because that science had not yet developed sufficiently standardized profiles of pedophiles and sexual psychopaths against which an alleged perpetrator could be matched. Also, *R. v. J.J*. is a case where the Court excluded evidence tendered by a psychiatrist based on testing with a penile plethysmograph to show lack of disposition of the accused to commit the offences with which he stood charged.

[29] In these kind of cases, the proffered evidence sometimes borders dangerously on evidence of propensity, character or even credibility as it did in *R. v. Marquard*, [1993] 4 S.C.R. 223. The trial judge must grapple with the importance of affording the accused a complete defence and the reluctance to exclude expert defence evidence on the basis of a cost/benefits analysis. These considerations do not apply here, but serve to illustrate the tension between the exclusion of relevant evidence and the need to protect the trier of fact from inherently unreliable evidence.

[30] A threshold reliability assessment has also been applied in civil cases. Loblaws relies on *Wolfin v Shaw*, [1998] B.C.J. No. 5 (B.C.S.C.) (QL) where Dillon J. excluded the evidence of a psychiatrist who sought to confirm a clinical diagnosis of mild traumatic brain jury based on the results of a PET scan. In *Wolfin*, the evidence was inadmissible because the technology had not been generally accepted for this purpose. However, this was a jury case. *McGuire v. Risling*, [2001] S.J. No. 684 (Sask. Q.B.) (QL) was also a jury case where brain-mapping evidence was excluded. Finally, I was referred to *Homolka v. Harris*, [2002] B.C.J. No. 831 (C.A.) (QL) a jury case where engineering evidence was excluded.

[31] It is difficult to find a civil case tried by judge alone, where novel scientific evidence was excluded because it failed to meet a threshold test of reliability. The reason may be that the metaphor of the judge as gatekeeper loses much of its symbolic force when it is the judge who is the trier of fact. This is not to say that a trial judge is excused from scrutinizing evidence as improperly admitted evidence can surely have an impact on a trial, but the likelihood of a judge being overwhelmed by the "mystic infallibility" of the evidence and misusing the evidence to distort the fact-finding process, is far more remote. The dangers that the principles are designed to avoid begin to fall away.

[32] None of the dangers that inform the cost/benefits analysis in *Mohan* are present here. Trial efficiency is not engaged as I have heard the evidence. There is little potential that I will be misled or overwhelmed by the evidence. There is little risk that the fact-finding function of the trial will be distorted. The evidence has little if any prejudicial effect on the trial process or on my role as the trier of fact.

[33] In this case, the plaintiffs' expert and the defendants' expert agree that there can be a relationship between trauma and PPS. Dr. Trojan's study and the studies relied on by Dr. Halstead and Dr. Bruno are small samples, but it is important to remember that Dr. Cashman's opinion is primarily based on his training, his education and his clinical experience with some five thousand PPS patients. If this is "novel scientific theory", it appears to be an accepted theory within the community that has knowledge of and experience with PPS. General acceptance of a theory is not a necessary criterion for admissibility, but it is one of the important factors because it offers some assurance that the theory is reliable. The ultimate reliability of Dr. Cashman's opinion has yet to be determined, but I conclude that it would be wrong to exclude it. Its probative value clearly outweighs any prejudicial effect and it therefore meets a threshold test of reliability.

Necessity in Assisting the Trier of Fact

[34] In *Mohan* at page 23, Sopinka J. states that he would not judge necessity by too strict a standard. He approves a passage from *Kelliher (Village of) v. Smith*, [1931] S.C.R. 672 at 684

that, “[t]he subject-matter of the inquiry must be such that ordinary people are unlikely to form a correct judgment about it, if unassisted by persons with special knowledge.” PPS, its risk factors and its relationship to trauma are outside common knowledge and experience. Dr. Cashman’s evidence satisfies this requirement.

[35] As with relevance, the necessity criterion is assessed in light of its potential to distort the fact-finding process and inherent in the application of this criterion is the concern that experts not be permitted to usurp the functions of the trier of fact: *Mohan* at 24. These concerns were the foundation of the rule that excluded expert evidence on the ultimate issue, although the rule is no longer of general application.

[36] Dr. Cashman and Dr. Bruno have different explanations for the onset of PPS symptoms that followed Dr. Chan’s traumatic accident. Dr. Cashman’s opinion approaches the ultimate issue in this trial, but it would be anomalous to exclude the evidence of Dr. Cashman because he concludes that there is a causal connection and admit the evidence of Dr. Bruno who concludes that there is not. This important issue cannot be determined without considering all of the evidence that bears on it. The importance of the evidence must be balanced against its potential to distort the fact-finding process. For reasons already given, this is unlikely to occur as I am capable of assessing the evidence and giving it the weight it deserves. The evidence therefore meets the necessity criterion.

Liability

[37] Dr. Chan’s accident occurred on August 20, 1993 in the Loblaws Superstore in the Erin Mills Town Centre when he slipped and fell on leakage from a watermelon bin. Graham Conroy, the Store Manager, observed the accident from his office. When he came to Dr. Chan’s assistance, he saw a puddle of liquid around the base of a wooden pallet or skid that held a container of watermelons. The container in which the watermelons were displayed was made of corrugated cardboard and was placed about ten feet from the store entrance at the edge of the produce area, near to the photo finishing counter.

[38] The accident occurred around 7:15 p.m. Mr. Conroy testified that cleanliness was a top priority in the store and that he or his staff carried out inspections every 30 to 45 minutes. However, on the day of the accident, the Sweep Log records only two routine cleanings of the produce area at 12:30 p.m. and at 6:45 p.m.

[39] Luigi Marcucci, the Assistant Produce Manager, was responsible for regular inspections of the produce area. He recalled mopping the area at 6:45 p.m. and finding that the watermelon bin was leaking. He estimated there were about 60 melons in the bin. As it was impractical to remove all of the melons to find the bruised or cracked watermelon, he cleaned around the area, waited a few minutes to see if the leak would recur, and then left. Dr. Chan slipped and fell soon after in the same spot where the leak had occurred.

[40] Dr. Chan's family were ahead of him and did not see him fall, but Mr. Conroy observed him walking at a normal pace before his fall. At that time, Dr. Chan wore a right long leg brace and walked with a cane held in his left hand. There is no evidence that he was hurrying or walking other than in his normal manner. He did not see the liquid until after his fall and described it as clear and sticky.

[41] The Accident Report describes the cause of the accident as "water leakage from watermelon bin". Although Mr. Conroy thought the liquid was coloured, he was prepared to rely on Mr. Marcucci's recollection that it was "very, very pale". Their estimate of the size of the puddle ranged from three to eight inches in diameter. Mr. Conroy and Mr. Marcucci said it was visible to them, but they are trained in spotting hazards of this kind and were looking for an explanation for the accident. I find that it would have been difficult for a shopper to notice a small puddle of very pale or colourless liquid pooling around a bin in that location.

[42] The *Occupiers' Liability Act*, R.S.O. 1990, c. O.2., s. 3(1), imposes an affirmative obligation on an occupier to take reasonable care for the safety of people who come on to its property. The standard of care is one of reasonableness in the circumstances and as circumstances vary from case to case, so do the requirements of meeting the statutory standard.

[43] Counsel referred me to relevant authorities which turn on their own particular facts, but the principles are not in question. Occupiers are not expected to be insurers, responsible for all harm that befalls visitors, but as an occupier has an affirmative duty to take care for the reasonable safety of visitors, this sometimes requires affirmative action. The duty to take reasonable care in the circumstances depends on what the occupier ought to foresee as being likely to occur. High-risk areas, such as produce departments, demand greater vigilance and better systems as the risk of spillage and leakage in these areas is greater, increasing the likelihood of injury: Allen M. Linden, *Canadian Tort Law*, 7th ed. (Markham: Butterworths, 2001) at 675 and 678-679; G.H.L. Fridman, *The Law of Torts in Canada*, 2nd ed. (Toronto: Carswell, 2002) at 586-587.

[44] Loblaws was aware that leakage and spillage is responsible for the majority of accidents that occur in its stores and that a produce department is a particularly vulnerable area. Their employees are trained to use floor mats in problem areas, such as where spraying occurs, to monitor floor areas regularly and to use signs where there are possible hazards. Although there appears to have been a regular inspection, maintenance and cleaning system in place at the Erin Mills store, once the leak was discovered, no other precautions were taken. A regular inspection and cleaning system does not meet the standard of reasonableness when the occupier knows that there is a risk of a recurring hazard that can lead to an accident.

[45] I find that Dr. Chan took reasonable care for his own safety and the slip and fall was caused by the negligence of Loblaws. Unless the damaged fruit was removed, it was foreseeable that further leaking would occur from a container made of corrugated cardboard with no waterproof liner. It was foreseeable that a small puddle of virtually colourless liquid on the floor would create a hazard for shoppers. If it was impractical to remove the watermelons to locate the source of the leak, Loblaws should have placed a warning sign or rubber mat near the bin. These simple measures could have prevented the accident. Loblaws accepted the risk of an accident and failed in its duty to take reasonable measures in these circumstances for the safety of Dr. Chan.

<u>Causation</u>

[46] I turn then to the important issue of causation. The governing authority is *Athey v Leonati*, [1996] 3 S.C.R. 458. The defendants will be held responsible if their negligence either caused Dr. Chan's injury or is a material contributing factor. Loblaws' negligence caused Dr. Chan's slip and fall that resulted in the injury to his left knee. It is responsible for these damages. The more difficult question is whether it is also responsible for Dr. Chan's present condition. Before turning to this evidence, I wish to comment briefly on my impression of the three witnesses who expressed an opinion on this.

Dr. Lipson

[47] Dr. Lipson described himself as a "non-expert expert" on PPS and was candid to acknowledge the limitations of his expertise in this area. He has treated many polio survivors, but only three patients with PPS and none who had developed PPS following trauma. He acknowledged that Dr. Cashman and Dr. Bruno were more expert in this area. Otherwise, I found Dr. Lipson to be a fair and credible witness in all respects.

Dr. Cashman

[48] Dr Cashman is a specialist in Internal Medicine and Neurology and an internationally recognized researcher and clinician in neuromuscular disease who has consulted on patients around the world. In 1998, he came to Toronto as Director of the Neuromuscular Clinic at Sunnybrook & Women's College Health Sciences Centre and holds the rank of Professor at the University of Toronto. His research has been published in leading medical journals and he has authored numerous abstracts, articles and textbook chapters on PPS as well as other kinds of neuromuscular disease. He has impressive credentials and he was an impressive witness.

Dr. Bruno

[49] Dr. Bruno trained in psychology and brain anatomy and has used this training to develop some of his theories on PPS. He was tendered as an expert in the research, diagnosis, etiology

and treatment of “postpolio sequelae”, including postpolio syndrome. He prefers the term, “pospolio sequelae” because he does not believe that PPS is a disease, although most physicians do. His qualifications were not challenged, but he was extensively cross-examined on the contents of his curriculum vitae. Regrettably, Dr. Bruno’s curriculum vitae embellishes his education and experience and presents a somewhat exaggerated and misleading picture of his credentials.

[50] Dr. Bruno has been a tireless and effective advocate and lobbyist on behalf of polio survivors, but he is not “the world’s leading expert on PPS” as he is described on the book jacket of *The Polio Paradox*. He is not a physician and the medical staff in New Jersey did not assess Dr. Chan. Dr. Bruno does not have the same qualifications, training, breadth of experience or credibility as Dr. Cashman and where their evidence conflicts, I prefer the evidence of Dr. Cashman.

Why did Dr. Chan develop PPS?

[51] There are three possible explanations for Dr. Chan’s present condition. First, that Dr. Chan had PPS before the slip and fall and this would have progressed regardless of the fracture. This is Dr. Bruno’s view. Second, that Dr. Chan was able to rehabilitate himself after the accident and that the symptoms that began to appear in 1995 and 1996 emerged independently of the fracture. This is Dr. Lipson’s view. Dr. Bruno also expressed this opinion. Third, that Dr. Chan was in a stable neuromuscular condition before the fall, but the fracture, immobilization and physiotherapy that followed his accident, precipitated the onset of PPS or contributed in a material way to the disease. This is Dr. Cashman’s view.

Did Dr. Chan have PPS before the 1993 accident?

[52] It is not controversial that after recovery from polio, there are a very small number of cells doing the work that a large number of cells did before the polio. PPS can be an insidious disease and the symptoms can onset gradually. According to Dr. Bruno, Dr. Chan was his own

risk factor for PPS due to his many years of overuse of polio-weakened muscles. Dr. Cashman strongly disagreed that Dr. Chan was bound to develop PPS.

[53] Before the accident, Dr. Chan had few, if any health problems. This is evident from the clinical notes and records of Dr. Peter House who became Dr. Chan’s family physician in 1986, seven years before his fall. Dr. House observed no signs of neuromuscular change or declining function until after the accident. Neither did Dr. Theo van Rijn, a physiatrist who has known Dr. Chan for many years. Dr. van Rijn’s pre-accident records were not available, but he did not recall Dr. Chan having any functional problems before his accident.

[54] Fourteen months before the accident, Dr. Chan experienced an acute episode of low back pain. Dr. Bruno testified that this was a prodrome or indicator of PPS that pointed to declining function. Dr. Bruno believes that polio survivors should “conserve to preserve” and is an advocate for the early use of assistive devices such as a wheelchair. He was critical of Dr. Chan’s treating physicians for failing to recommend this to Dr. Chan.

[55] Dr. Chan had postural and gait abnormalities that increased the lordosis in his lower spine, but he had walked for years in the same manner without apparent difficulty. His physicians saw no reason to recommend to him that he use a wheelchair or reduce his level of activity and neither did Dr. Cashman. I prefer their evidence.

[56] John Oldham is a physiotherapist who examined Dr. Chan at the time of the low back pain episode. He made some suggestions to help him with his posture and to improve his gait. He did not see Dr. Chan again until after his accident. Dr. Chan did not see a physician or other health care professional for any reason until after the accident.

[57] The consistent evidence from Dr. Chans’ physicians, professional colleagues and family is that Dr. Chan enjoyed excellent health. He was a highly energetic individual who was fully engaged in a very busy professional and personal life. He worked extraordinarily long hours, between 60 and 75 hours a week. He drove a car. He travelled. He raised a family. He was a frequent and gracious host to family and friends. He enjoyed hobbies such as painting and

regular half hour evening walks with his wife. He functioned at a very high level. There is no medical or other evidence of declining function before the accident. This is not a picture of an individual with emerging PPS.

[58] Dr. Cashman and Dr. Chan's treating physicians reject the idea that Dr. Chan had PPS prior to August 1993 or that it would have inevitably emerged. Dr. Lipson agrees that it cannot be established that PPS started at an earlier time. Dr. Bruno is alone in the view that Dr. Chan had early signs of PPS, but there is no credible evidence to support this. Before the accident, Dr. Chan was a vigorous, healthy individual. The low back pain he experienced in June 1992 was a short-lived, isolated and unremarkable event. I find that Dr. Chan did not have any indicators of PPS before the accident.

Did Dr. Chan recover from the knee injury?

[59] Dr. Chan expected to recover from his injury and he put considerable effort into achieving this. Loblaws submits that by and large, he did achieve this and that he had significantly recovered from his knee injury before PPS symptoms of fatigue, lack of energy and pain began to emerge in 1995 or 1996.

[60] In July 1995, Dr. Chan's treating orthopaedic surgeon, Dr. Brian Day, wrote that Dr. Chan had shown only "minimal improvement" and that he has "persistent symptoms that have plateaued". Dr. Lipson testified that he relied on Dr. Day's opinion to conclude that Dr. Chan was able to slowly recover some of his strength and had some increase in function. He thought that the severe loss in strength after this was not diagnostic of atrophy of disuse, but of loss of function gradually over time and was best explained by PPS.

[61] Dr. Lipson based his diagnosis of PPS on Dr. Chan's complaints of extreme fatigue, his clinical observations of severe atrophy of the muscles of the left leg and, importantly to Dr. Lipson, the absence of left ankle dorsiflexion. Dr. Lipson noted that left ankle dorsiflexion was present on Dr. van Rijn's examination in September 1996, but when Dr. Lipson examined Dr. Chan in August 2002, this was completely absent. The major dorsiflexor of the foot is the tibialis anterior muscle, which originates in and around the knee from the lateral condyle of the tibia. The absence of dorsiflexion (the ability to lift the foot up) indicated to Dr. Lipson that some new neurological weakness had occurred, which is one of the important diagnostic criteria for PPS. This is the principal basis of his diagnosis and of his opinion that PPS emerged after the accident, but independently of it.

[62] Disruption of the knee joint by fracture and subsequent immobilization could negatively impact the tibialis anterior muscle. Dr. van Rijn testified that Dr. Chan has limited use of the tibialis anterior muscle, but has always had the use of the toe extensor muscles that dorsiflex the ankle, although they are now weaker. When Dr. Cashman examined Dr. Chan in November 2003, left ankle dorsiflexion was present to a similar degree as it had been in 1996 (Dr. van Rijn found grade 2 or 3; Dr. Cashman found grade 2).

[63] In 2001, Dr. Chan was fitted for a long leg brace on his left leg similar to the one he has always worn on his right leg. There then followed a long, time-consuming process to fit this brace and it continues to require constant readjustment. As part of this process, Dr. Chan went on to develop problems from the brace itself and lost dorsiflexion of the ankle, but this was not permanent. This was corrected when the ankle and foot component of the brace was removed. Dr. House notes this in his clinical record in June 2002 and in October 2002 records, “he now has some return of dorsiflexion”. By the time Dr. Chan saw Dr. Cashman a year later, it would appear that this had improved further and was about the same as it had been three years earlier. The dropped foot that Dr. Lipson observed in August 2002 was related to the bracing problems and not to new neurological weakness. Dr. Lipson’s diagnosis was correct, but the basis for it is incorrect.

[64] Although Dr. Bruno and Dr. Lipson agreed with Dr. Cashman that trauma can be a risk factor for PPS, they deny the causal relationship in Dr. Chan’s case. I have already explained the principal basis for Dr. Lipson’s opinion. Dr. Bruno based his opinion on evidence that led him to conclude that Dr. Chan had significantly recovered from the knee injury about 34 months post-accident. He relies on three main indicators: Dr. Chan was walking again with a cane by June 1996, he was working 55 hours a week by that summer, and he regained strength in his left quadriceps.

[65] Mr. Oldham described Dr. Chan before the accident as having weak musculature in the left lower extremity and Grade 2 function in the left quadriceps muscle. However, he had sufficient control of the left knee extensor and flexor muscles and the quadriceps muscle to permit him to stabilize the left knee, weight bear on his left leg and using the support of the cane and the musculature in his lower back, to rotate the pelvis and bring the braced right leg through. According to Mr. Oldham, Dr. Chan had a well-functioning left knee joint.

[66] After the accident, Dr. Chan attended physiotherapy twice-weekly with Mr. Oldham for two years. Dr. Chan was also exercising at home. Nevertheless, he never recovered function in the left quadriceps muscle. According to Mr. Oldham, there were times when he was able to get a flicker of quadriceps function (Grade 1), but he ran into problems as soon as he started to

weight bear. Mr. Oldham's evidence is consistent with the observations of Dr. House in October 1995 that the left leg was "very atrophic". More than two years after the accident, he was observing the same loss of muscle bulk that was present immediately following the injury and immobilization of the knee joint. The following examples from the clinical record of Dr. House illustrate this:

> March 1994, left knee discomfort, can't drive yet, using 2 crutches; July 1994, can walk with crutch Lt. arm & cane Rt side, cannot drive; October 1994 (Dr. Day), significant left lower extremity weakness; January 1995, ongoing L knee problem, can walk max. 3 blocks due to pain & fatigue, tried driving but increased pain in L knee; Sept. 1995, L knee pain persists, driving short distances, pain at night, quadriceps weakness+++; October 1995, L leg very atrophic; June 1996, Lt knee persistent pain; October 1996, walking up an incline, left leg gave way; Sept 1996 (Dr. van Rijn) aching in L knee and tripping, fatigues more easily, all in all, his function has decreased; October 1996, driving short distances, walking very short distances, Max 4 blocks, previous 10-20 blocks; May 1997, can only walk 3 blocks slowly, can only climb with rails, had to have elevator installed in home, more pain; December 1998, left knee swelling and weak, since accident if tries to walk with right knee brace and cane in left hand develops more pain and swelling; February 00, still very weak left knee.

[67] Loblaws relies on other indicators of improvement, pointing out, for example, that Dr. Chan's remunerative hours in 1997 were actually greater than in 1992 and he attended more conferences after the accident than before. While this is true, he also worked twice as many days in 1997 as he had in 1992. Despite this, he was slow to produce reports. He became largely invisible to his professional community. Robert Charlton and Thomas Teed are personal injury lawyers in Vancouver who were referral sources for Dr. Chan. They describe a very different person after the accident. So do his family, his work colleagues and his art teacher, Danny Chen.

[68] There is no doubt that Dr. Chan attempted to resume the life he led before the accident, but I do not agree that he was successful or that the evidence supports a finding that he largely recovered. Certainly, the medical evidence does not support this. He resumed his working life, but this was at the expense of other activities that he previously enjoyed and that were important to his personal and professional life. He is a talented painter and continued with his art lessons

for a while, but fatigue and pain interfered. His office became his life and by and large, he devoted his energy to maintaining his practice. For a man who regularly worked up to 75 hours a week, a working week of 55 hours, three years post-accident, is not a return to a normal or a near normal level of activity.

[69] Through determination and hard work, characteristic of those who are polio survivors, Dr. Chan was able to rehabilitate himself to an extent and to resume some level of activity, but he never returned to his previous level of health or vigour. The evidence as a whole reveals a picture of an individual struggling to regain function, rather than of a person who recovered. I find that Dr. Chan did not significantly recover from the injury to his knee.

Did the accident cause or contribute to PPS?

[70] Following an accident and a period of immobilization, muscle atrophy occurs because the muscles are not contracting. In a person who has not been affected by polio, there is a normal pool of motor neurons and motor units to bear the brunt of physiotherapy to increase muscle mass during the rehabilitation process. However, for a post-polio patient, unaccustomed exertion can precipitate new weakness. Dr. Cashman describes this as “the straw that breaks the camel’s back of these highly stressed motor nerve cells and their axonal trees”. Dr. Bruno describes this as “blowing a fuse” and writes, “when fuses blow, neurons work less well and muscle weakness, fatigue and pain may follow”. [Ex. 76]. Because polio survivors have fewer cells to work with that are in a vulnerable state, overworking the polio-weakened neurons can make things worse. This explains why trauma can be a risk factor for PPS.

[71] Dr. Lipson rejected the link between trauma and PPS in Dr. Chan’s case because he thought that if there had been a disease process going on between 1993 and 1995, this would have interfered with the healing that did take place. But what healing did take place? By July 1995, two years post-accident, he had shown only minimal improvement and had persistent symptoms that had plateaued. This means he was neither getting better, nor worse. By September 1996, Dr. van Rijn noted that his function had decreased. According to Dr. Cashman, whose opinion I accept, any improvement in function between 1993 and 1995 was due to the

rehabilitation for his patellar injury and not due to any delay in the onset of PPS symptoms. The clinical records of Dr. House, Dr. van Rijn, Dr. Day and Mr. Oldham are consistent with this. None of the improvement was lasting. After Dr. Chan discontinued physiotherapy in the summer of 1995, his decline is even more evident.

[72] Post Polio Syndrome cannot be explained on the basis of new neurological weakness in the left ankle as Dr. Lipson thought. It cannot be explained on the basis that Dr. Chan would have developed PPS regardless due to a lifetime of overuse. He had no indicators of PPS before the accident. Not all polio survivors develop PPS and apart from having had polio as a young child, Dr. Chan had none of the known risk factors.

[73] The only risk factor was the accident. Dr. Chan had recovered well from paralytic polio. He was healthy. He had led an active life. He was completely stable on a neuromuscular basis and had been for more than forty years. In polio survivors, there is neuromuscular junction instability and motor unit remodelling that precedes the onset of symptoms, but this process can be balanced for decades as it was for Dr. Chan.

[74] I find that the most probable explanation for Dr. Chan's present condition is that the fall and resulting immobilization and treatment triggered the PPS that Dr. Chan now has. But for this event and subject to the risk he faced as a polio survivor (a limiting factor relevant to the assessment of damages), he likely would have continued to be stable as he was before. The plaintiffs have established causation on a balance of probabilities.

Material Contribution

[75] If I were to accept Loblaws' explanations for Dr. Chan's present condition, I would arrive at the same place on the basis of the material contribution test described in *Athey* and *Alderson v Callaghan* (1998), 40 O.R. (3d) 136. Before the accident, Dr. Chan's left leg functioned and permitted him to ambulate surprisingly well. His disability was practically invisible to those with whom he worked and socialized, but he had only one functioning leg. It was not however, a normal leg, making it more vulnerable and considerably more difficult to rehabilitate. A fully

functioning quadriceps muscle is Grade 5 whereas before the accident, Dr. Chan's left quadriceps was Grade 2.

[76] Dr. Martin Roscoe is an orthopaedic surgeon who assessed Dr. Chan at the request of the defendants in August 2002. Dr. Roscoe agreed that the abnormal biomechanical situation in Dr. Chan's left knee and the pre-existing atrophied musculature would lead to abnormal stresses on the left knee joint and at some point to degenerative changes, making ambulation very difficult and ultimately, impossible. He also agreed that for Dr. Chan, fatigue is a significant factor because of the sheer effort required to ambulate.

[77] Although Dr. Lipson thought that PPS developed independently of the fracture for reasons I have already explained, he agreed that the fracture was a contributing factor and aggravated the disease. Thus, even if the injury and its consequences did not cause PPS, this evidence establishes that the fracture and its resulting impact on Dr. Chan's ability to function as before is a material contributing factor to his overall impairment.

Assessment of Damages

Non-Pecuniary General Damages

[78] According to the plaintiffs, Dr. Chan's injury is catastrophic and similar to incomplete paraplegia that will progress to complete paraplegia. They submit that these damages should be assessed at $300,000, which is at the level of the current maximum prescribed by the *Trilogy*. Dr. Chan was 52 years of age at the time of the slip and fall and engaged in a full and vigorous life that he expected to continue well into his senior years. His father is alive and healthy in his 90's; his grandmother lived to 104.

[79] Dr. Bruno testified that Dr. Chan's PPS can be managed if he uses a wheelchair and "conserves to preserve". He has written that the usual course of PPS is deterioration at the rate of one to seven per cent a year. This is consistent with Dr. Cashman's evidence. I find that there is no cure or effective treatment for PPS and Dr. Chan can expect gradual decline. He will lose the ability to ambulate and there are already signs of this. He has had to adapt to a different life and

his loss of independence has resulted in changes in his personality and mood that have affected his interactions with family and friends.

[80] The defendants point out that Dr. Chan currently functions at a high level, working 40 hours a week at the age of 64 and complaining only of mild to moderate pain, which has not changed since the accident. In their view, his limitations do not preclude him from continuing with a satisfying professional and personal life. They propose an assessment of $150,000.

[81] In assessing damages, the court must attempt to restore the plaintiff as far as money can do so, to the position that he would have been in prior to the accident with all of its attendant risks and shortcomings and not to a better position: *Athey*. Pre-existing medical problems and a predisposition to develop the disability brought about by an accident should be considered in assessing general damages because the court must compare the plaintiff's anticipated life after the accident with the life the plaintiff would have enjoyed but for the accident: *Koukounakis v. Stainford* (1995), 23 O.R. (3d) 299 at 306 (C.A.). An assessment at the maximum would ignore the realities of Dr. Chan's life as a polio survivor and the risks this presented for him had the accident not occurred. An assessment at the amount proposed by the defendants would ignore the realities of the changes the accident has brought to Dr. Chan's life. I assess the non-pecuniary general damages at $185,000.

Family Law Act Damages of Betty Jung

(i) Section 61(2)(e)-Care Guidance and Companionship

[82] The *Family Law Act*, R.S.O. 1990, c. F.3 enables a family member to claim compensation for the loss of care, guidance and companionship of an injured plaintiff. The upper end of the range for this head of damage is currently about $90,000. The plaintiffs submit that Ms Jung's damages should be at the high end and that an award of $75,000 to $85,000 would appropriately affect her loss. The defendants submit that Ms Jung's damages should be assessed at $35,000.

[83] Betty Jung and David Chan have been married for thirty-one years and have enjoyed a fulfilling relationship and an enduring marriage. For many years, Ms Jung has worked as the office administrator in her husband's practice, but within their marriage, their roles are traditional. Dr. Chan's domain is the office and hers is the home. They support one another in each other's domain, but before the accident, Dr. Chan led an independent life. He is now more dependent on his wife. She has had to curtail some of her own activities to be available to him and their shared activities have diminished or disappeared. Their son, Stephen, describes them as less affectionate and loving with one another. Dr. Chan is often tired, frustrated and irritable and this has strained their relationship. Nevertheless, he remains a caring and considerate husband and she remains a devoted and supportive wife.

[84] I am satisfied that Betty Jung has suffered a loss, but I do not think that it attracts an award at the upper end of the range. The assessment of this kind of loss is difficult to quantify, but I find that an appropriate assessment is $50,000.

(ii) Cost of care

[85] At common law, there is compensation for the cost of care provided by a family member to an injured person: *Andrew's v. Grand & Toy Alta. Ltd.*, [1978] 2 S.C.R. 229, (1978), 83 D.L.R. (3d) 452; *Samples* at p. 8; *Brophy v. Turnbore*, [2000] O.J. No. 3020 at p. 12 (S.C.J.) (QL); *Granger v. Ottawa General Hospital*, [1996] O.J. No. 2129 (Ont. Ct. Gen. Div.) (QL) at para. 157. In Ontario, this overlaps with section 61(2)(d) of the *Family Law Act*. This subsection allows specified family members to claim for the value of nursing, housekeeping and other services.

[86] The plaintiffs ask me to quantify this loss using hourly rates for professional attendant care services provided by the costs of care experts. Neither expert expressed an opinion on the amount of time Ms Jung has spent caring for her husband and she kept no record of this, but there is evidence that she assists him on a daily basis, particularly with transportation. I am asked to base my finding on a "conservative" estimate of 10 hours a week over 12 years, producing an amount in the range of $120,00 to $150,000.

[87] In *Dubé (Litigation Guardian of) v. Penlon Ltd*. (1994), 21 C.C.L.T. (2d) 268 (Ont. Ct. Gen. Div.) at para. 193, Zuber J. noted that the estimate of these types of losses is more art than science and preferred to assess an amount for care-giving services on a conventional basis. Cunningham J. (as he then was) followed this approach in *Granger* and assessed the amounts of $50,000 and $75,000 respectively as being appropriate to compensate Mrs. Granger and Mr. Granger for their services to their severely brain-injured daughter over six and one half years.

[88] In *Samples*, the result of using an hourly rate multiplied by the time Mr. Samples had spent caring for his wife (in that case, supported by expert evidence that Mr. Samples was devoting six hours a day to her care), Carruthers J. observed that this produced an amount that was twice as large as the monthly amount for professional “live-in” help Mrs. Samples would require in the future. He found this to be an unreasonable amount to require the defendants to pay.

[89] In *Bain (Guardian ad litem of) v. Calgary Board of Education*, [1993] A.J. No. 952 (Q.B.) (QL), the plaintiffs put forward a calculation on an hourly basis, which resulted in a figure of about $20,000, but the defendants did not dispute the amount or the basis of the calculation. Virtue J. found it to be a moderate amount that was fair to both parties and represented the loss sustained by Mrs. Bain in caring for her injured son.

[90] I was referred to *Brophy v. Turnbore*, [2000] O.J. No. 3020 (S.C.J.) (QL), where Mr. Brophy was rendered a quadriplegic by an accident. Scime J. assessed the value of services provided by family members at $70,000 for his wife and smaller amounts for his two sons. In *Brophy*, there was opinion evidence valuing the cost of care, but the Court observed at para. 51 (echoing the comments of Zuber J. in *Dubé*) that care-giving services are “incapable of strict arithmetical calculation”.

[91] Ms Jung is entitled to recover for the value of the services that she has provided to her husband over the past 12 years. Her services have not been as extensive as those of the family members in the cases to which I have referred. Accepting that this is more art than science and

that I should attempt to arrive at a reasonable amount that is fair to both parties, I assess damages for these services at $60,000.

Special Damages

[93] Dr. Chan worked very hard his entire life to avoid being perceived as disabled. He was initially very optimistic about his recovery and it was some years before he realized he was not getting better. This required an understanding and acceptance of his limitations, which is not an easy process for anyone. Nonetheless, in January 1994, at a time when he expected to fully recover from his injury and long before he was diagnosed with PPS, Dr. Chan and his wife decided to build a new home that incorporated a number of features to assist Dr. Chan with his mobility. They bought a lot and built a multi-level house with an elevator. Ms Jung testified that they would have preferred to build a single-level home, but this required a ¼ acre lot, which was difficult to find in the city of Vancouver.

[94] At trial, the plaintiffs attempted to introduce evidence of the cost of the new home through Betty Jung. The defendants objected as no expert report was delivered, although the plaintiffs' former counsel had given an undertaking to do this if the claim was pursued. I ruled that the defendants were prejudiced and did not permit this evidence to be adduced. As a result of my ruling, the claim was confined to the cost of the elevator and small amounts for modifications to the home.

[95] Before the accident, Dr. Chan was able to go up and down stairs. After the accident, he could not do this. In my view, this was as a result of the patellar fracture and the inability to rehabilitate his knee due to his pre-existing condition. Thus, even if PPS had not emerged, he would have been required to install an elevator in his home. The modifications to the bathroom and kitchen are for modest amounts and are appropriate.

[96] Exhibit 48 was filed as a summary of the agreed amount for special damages as of June 13, 2005, subject to my findings on liability and causation. The amount is $57,027.23. I exclude item number 13 in the amount of $455.82 as this expense relates to payment to an architect for a permit for the driveway in the new home. I find the remaining items to be reasonable and appropriate items for medical and rehabilitation expenses. The total amount is $56,571.41.

[97] A summary of one-time costs was prepared by the plaintiffs' cost of care expert, Lila Quastel, and filed as part of Ex. 38. Ms Quastel is an occupational therapist and served as Assistant Professor of Rehabilitation in the Faculty of Medicine, University of British Columbia between 1971 and 1990. For many years, she also served as a long-term care consultant in the public sector. Since 1991, she has consulted privately. By and large, Ms Quastel's evidence was challenged only on the issue of causation.

[98] Ms Quastel's summary includes amounts for assistive and adaptive aids, the most substantial item being a van with a lift and hand controls. The plaintiffs' written Outline of Closing Argument does not refer to the items listed on exhibit 38 as part of the claim for special damages, but I believe this is an oversight. I propose to make findings on this evidence as I am confident that counsel for the plaintiffs will advise me if the claim for Dr. Chan's special damages is limited to the agreed amount on Ex. 48.

[99] I am satisfied based on all of the evidence that Dr. Chan requires the aids listed on Exhibit 38. I allow the items in the amounts claimed, with these exceptions. I have excluded the amounts of $23,874 for elevator installation and $1,200 for the in-home scooter as these items are duplicated on Ex. 48. The medical evidence does not establish that orthopaedic shoes are related to the accident. I have not allowed this item. I am not satisfied that Dr. Chan required an office assistant for one year. I have not allowed this item. I have allowed the van in the amount of $52,500. The total amount for the items I have allowed is $96,858.

[100] I assess Dr. Chan's special damages in the total amount of $153,429.41 ($56,571.41 + $96,858).

[101] The agreed amount for the subrogated claim of the British Columbia Ministry of Health is set out on Ex. 49 and is $3,699.59.

Future Economic Loss

Burden of Proof

[102] The evidentiary burden for future economic loss is lower than the balance of probabilities. If the plaintiff establishes a real and substantial risk of future pecuniary loss, he is entitled to be compensated: *Shrump v. Koot* (1977), 82 D.L.R. (3d) 553 at 556-9 (Ont. C.A.). The measure of compensation for future economic loss will depend on the possibility, if any, that the plaintiff would have suffered some or all of the loss, even if the wrong had not occurred: *Graham v. Rourke* (1991), 74 D.L.R. (4th) 1 at 13 (Ont. C.A.).

Contingency

[103] Contingencies are factors affecting the degree of risk of future pecuniary loss and recognize the possibility that some of those losses may have occurred in any event. There are both general and specific contingencies. General contingencies are common to the future experience of all of us and are not readily susceptible to evidentiary proof and may be considered in the absence of such evidence. A trial judge may adjust an award by a modest amount to give effect to general contingencies, but an allowance for a specific contingency must be supported by evidence of the real possibility of the contingency occurring: *Graham v. Rourke* at 14-15.

[104] As a polio survivor, there was a risk that Dr. Chan would suffer from PPS even if the accident had not occurred. Some years ago, it was thought that about twenty-five per cent of polio survivors would develop PPS. This was the contingency factor used in *Samples*. Dr. Cashman and Dr. Bruno agree that the risk is higher and that between fifty and fifty-eight per cent of polio survivors will go on to develop PPS. As Dr. Chan had no risk factors for PPS apart from having had polio, Dr. Cashman thought that the risk factor for Dr. Chan was somewhere between zero and fifty per cent and likely, down towards the lower end of this range. The defendants did not cross-examine Dr. Cashman on this aspect of his evidence.

[105] The plaintiffs submit that given Dr. Cashman's uncontradicted evidence, a fair contingency to apply to future economic loss is in the range of ten to fifteen per cent. The

defendants did not make submissions on the contingency factor to apply. I accept the plaintiffs' submission and find it appropriate to apply a contingency factor of fifteen per cent to the future economic losses.

Future Care

[106] The defendants' cost of care expert was Lucinda Soini. She trained as a social worker and has many years of experience assessing the future needs and care costs for persons who have disabling injuries. Ms Quastel had the advantage of assessing Dr. Chan in his home in 2002 and 2004.

[107] Ms Quastel and Ms Soini came to divergent opinions, mainly because Ms Soini prepared her report and assessment of cost [Ex. 92A] on the basis that Dr. Chan's PPS is unrelated to the accident and she did not allow for many of the items that Ms Quastel thought would be required. In view of my finding on causation, I prefer Ms Quastel's evidence. I am satisfied that there is a real and substantial possibility that Dr. Chan will require the aids Ms Quastel described in her evidence and are found on Ex. 38.

[108] Ms Quastel estimates the annual cost of future care at $64,000. I have excluded the following items as the evidence does not support them: swimming admission fee, swimming attendant care, horseback riding, custom made orthopaedic shoes replacement, clothing allowance. I have allowed the van replacement in the amount of $6,500. I have excluded the amount for attendant care/homemaker, which I will address separately below. The total annual cost of the items I allow is $22,010.95.

[109] Ms Quastel included provisional costs in the event Dr. Chan requires surgery on his left knee. The medical evidence does not support surgical intervention and I have not allowed any amount for this.

Attendant Care

[110] Ms Quastel included the provisional amount of $91,250 for a full-time live-in attendant caregiver on the assumption that Ms Jung is not available to assist Dr. Chan. She also included an amount for part-time attendant care four hours a day on the assumption that Ms Jung will be there to provide assistance. Ms Quastel used an hourly rate of $22.95 per hour for fifty-two weeks resulting in an annual amount of $33,507.

[111] Ms Soini included an allowance for 10 hours of care/homemaking assistance per week in the event that Ms Jung is not there to assist her husband. She used an hourly rate of $18.00 per hour for forty-eight weeks resulting in an annual amount of $8,640.

[112] The plaintiffs suggest that I approach this on the basis that Ms Jung, who is eleven years younger than her husband, will be available to provide assistance to him. They suggest that I take the difference between the cost of a full-time live-in attendant caregiver and the cost of part-time attendant care. This produces an annual cost of $57,743.

[113] Currently, Dr. Chan is physically independent in personal care. Except on a very occasional basis, he is not involved in meal preparation or housekeeping and was not involved in this before the accident as this is Ms Jung’s domain. As I have said, the normal course for PPS is deterioration at the rate of one to seven per cent per year. I reject Dr. Bruno’s evidence that this can be managed by using mobility aids. Dr. Lipson and Dr. Cashman agree that Dr. Chan can expect to decline.

[114] My impression is that Dr. Chan will be unable to ambulate with forearm crutches for very much longer. He already uses motorized transport at home and when he travels. Once he is in a wheelchair, he will require more assistance than he does now. When this will occur and how much more assistance he will require is difficult to predict.

[115] Doing the best I can with this evidence, I find that Dr. Chan will require attendant care four hours a day and that this will be provided either by a paid professional or Ms Jung, or a combination of both. I have taken into account the contingency that Ms Jung will not be available to assist him, but I think the more likely scenario is that she will. I see no reason to

limit attendant care to 48 weeks a year. I find an appropriate hourly rate is $20.00. This produces an annual amount of $29,200.

[116] I find the total annual care cost is $51,210.95 ($22,010.95 + $29,200).

[117] The parties agree on the present value factors calculated as at June 30, 1995 by Gary Principe filed as Ex. 52. Dr. Chan's future life expectancy is 14.072 years. Applying this present value factor, I find the amount for future care is $720,640.48.

Income Losses

[118] Before the accident, Dr. Chan operated two businesses from the same premises, Chan & Associates Consulting Inc. ("CASCI") and Integrated Rehabilitation Services Inc. ("IRSI"). IRSI mainly used sub-contractors who provided tutoring, counselling and other rehabilitation services to Dr. Chan's clients. CASCI provided psychology services and used a combination of sub-contractors and paid employees. The plaintiffs' position is that business declined after the accident because Dr. Chan was unable to do promotional work which would have generated new referrals due to lack of energy, focus and concentration. As well, he had to devote more time to assessments and the preparation of reports because he worked much less efficiently.

[119] The plaintiffs' damages expert, John Douglas, prepared calculations of economic loss from the combined operations of CASCI and IRSI on the assumption that the losses are attributable to Dr. Chan's injuries due to the slip and fall and that there were no outside factors that would have adversely affected the gross revenue and pre-tax income of CASCI/IRSI. The defendants' expert, John Siegel, challenged this. Specifically, he sought to demonstrate that entertainment, promotion and conference expenses declined only moderately in the five years after the accident, that gross revenues of CASCI were growing after the accident and that gross revenues of IRSI were already declining before the accident. There is evidence to support this.

[120] At the time of the accident, IRSI was receiving about 95 per cent of its revenue from the Insurance Corporation of British Columbia ("ICBC"), a provincial crown corporation established to provide universal compulsory automobile insurance to BC motorists. CASCI was receiving in

excess of 25 per cent of its revenue from this source. It seems that Dr. Chan was able to generate the majority of revenue for CASCI from private patients. After the accident, CASCI revenues were growing until fiscal year 2000 and then began to decline. IRSI revenues declined by more than $100,000 in the year after the accident, but then remained at a consistent level until fiscal year 1998 when there was a very sharp decline. In calendar year 1998, the business ceased to operate.

[121] The post-accident losses are almost entirely due to the decline in IRSI revenue. Mr. Douglas did not investigate what external factors may have caused this or why IRSI ceased operations entirely in 1998. He relied on Dr. Chan's explanation that as employees and sub-contractors left, he did not replace them as he lacked the energy to do this.

[122] The defendants advanced another explanation for the decline in IRSI revenue. Dr. Chuck Jung is a psychologist who was on the payroll of IRSI until the end of 1993 and then on the payroll of CASCI until the end of 1995. No reason was given for why this changed. When Dr. Jung left, he opened his own practice. The decline in Dr. Chan's revenue from ICBC generally corresponds to the growth in payments to Dr. Jung. The suggestion is that he took clients with him. Also, the decline in IRSI revenue in 1994 matches Dr. Jung's departure from IRSI to CASCI.

[123] I do not necessarily accept the defendants' explanation, but I am troubled by the plaintiffs' explanation and it is their onus to establish the past economic loss on a balance of probabilities. There is no doubt that in the years following the accident, Dr. Chan had to work very hard to maintain his practice, but he also expected to recover from his injuries. It does not make sense that he did not replace Dr. Jung or the other employees and sub-contractors as they left. The majority of the employees and sub-contractors at IRSI were tutors, life skills counsellors and vocational counsellors, who were working part-time or on short contracts. Betty Jung was the office administrator and far more involved in the daily administration of the business than Dr. Chan who devoted his time to the professional work of a neuropsychologist. It is reasonable to think that she, rather than Dr. Chan, would have looked after the hiring of staff. The plaintiffs' explanation does not satisfy me.

[124] Despite the lack of analysis or adequate explanation for the decline in IRSI revenue, which is the main reason for the income loss, the defendants concede that it is fair to assume that there was one. They dispute the methodology used by Mr. Douglas.

(i) Past Loss of Income

[125] In preparing the calculations of past loss of income, Mr. Douglas used a combination of simple and weighted averages and estimated the pre-tax income loss at $405,000. He employed a similar method for the calculation of post-accident income, which he estimated at $275,000. The difference is $130,000 per year. This produced a past loss of income in the amount of $1,296,700.

[126] Mr. Seigel applied a simple average of the four fiscal years before and after the accident. Using this approach, the average annual pre-accident income is $380,100 and the annual average post-accident income is $348,900. This produces a past income loss of $343,200 ($31,200 x 11 years).

[127] I prefer Mr. Seigel's approach to the calculation of pre-accident income because $380,100 corresponds with the average earnings of CASCI/IRSI in the six years pre-accident and with the average earnings in the four years post-accident. Mr. Douglas' approach gives too much weight to the two highest pre-accident income-earning years and includes the years 1988 and 1989 where significantly more was spent on promotion. I accept Mr. Seigel's opinion that this indicates that Dr. Chan ran his business differently in those years and these years should not be included in the pre-accident average.

[128] Mr. Seigel was criticized for using a four-year average to calculate post-accident income. A simple average for all of the post-accident years produces a number that is higher than Mr. Douglas' weighted average of $275,000, but lower than Mr. Seigel's simple four-year average of $348,900.

[129] In the four post-accident years (1994, 1995, 1996 and 1997), CASCI revenue increased producing an average income of $415,564. IRSI revenue declined in 1994, but then remained at a

consistent level for the next three years. There was a significant decline in 1998 as the income in that year was $98,541 compared with $214,159 the previous year. IRSI produced no income after 1999 and in that year, the income was only $5,617. The post-accident four-year average for IRSI income is $247,062.

[130] If CASCI revenue is considered on its own, it is clear that there is no demonstrable post-accident loss of income. Mr. Seigel's approach gives some recognition to the IRSI losses, which arguably should not be included at all. Mr. Douglas' approach overestimates the post-accident loss as IRSI earned no revenue in the years 2000, 2001, 2002 and 2003 and significantly less revenue in the years after the accident than before for reasons that were not independently investigated and have not been satisfactorily explained. In my view, Mr. Seigel's approach more accurately reflects the income losses. I find that the past loss of income is $343,200.

(ii) Future Loss of Income

[131] Mr. Douglas calculated the future loss of income on the basis that Dr. Chan would work to age 70 and that his productivity should be measured in relation to his expected decline in function of one to seven per cent per year. In doing this, Mr. Douglas assumed that there is a direct relationship between the degree of impairment and the decline in revenue. In cross-examination, he agreed that there is no such necessary relationship. The plaintiffs' position is that I should accept Dr. Chan's evidence that he would have worked to age 70 and that I use the midpoint of the expected range of deterioration, which is 3.5 per cent. Using this approach and the estimated loss of income calculated by Mr. Douglas, the present value of the future loss of income for CASCI/IRSI is $1,074,637. According to the plaintiffs, a contingency of 1 per cent per annum should be applied to this based on Dr. Cashman's evidence that healthy people will decline at this rate as they age.

[132] Many self-employed professionals today work past the age of 65. I accept Dr. Chan's evidence that he planned to work to age 70. The main factor that would affect Dr. Chan's ability to earn income in the future is fatigue. Dr. Chan is currently 64 years of age and is working 40 hours a week. I believe he would have been able to work longer hours if the accident had not

occurred. However, between the ages of 64 and 70, I do not believe he would be working more hours than he does now. As Dr. Cashman testified, his capacity for work would decline even if the accident had not occurred. I have already found that Dr. Chan's past loss of income is $31,200 a year. A contingency of 1 per cent per year for seven years reduces this to $29,016. Applying the present value factor of 5.0988 to age 70, I find that the amount for future loss of income is $147,946.78.

(iii) The Pain Clinic

[133] The main purpose of Dr. Chan's visit to Toronto in August 1993 was to attend a two and one half day workshop presented by The Coping Skills Institute on how to develop a coping skills program for individuals with chronic pain. Dr. Chan testified that had the accident not occurred, he planned to add a pain clinic to his portfolio of services.

[134] Mr. Douglas projected the loss of income from the pain clinic to be $2,570,505 from information provided by Dr. Chan and Betty Jung. The proposed clinic was to be a psychology-based clinic, rather than one based on a multidisciplinary model. There are at least two pain management clinics in Vancouver, but both operate on a multidisciplinary model. There is no pain clinic in Canada of the kind proposed by Dr. Chan. Mr. Charlton and Mr. Teed thought that there was a need in Vancouver for more services for individuals with chronic pain, but they assumed that Dr. Chan's clinic was based on a multidisciplinary model.

[135] The Coping Skills Institute is based in Lafayette, Louisiana and appears to be a business initiative of Jimmie Cole and his wife, Molly. He is a clinical psychologist and she is a therapist and teacher. They lead workshops and have developed a set of materials that they offer for sale. The promotional material includes this statement: "You will also learn effective billing for workers' comp., Medicare, Managed Care and Group Insurance". I infer that pain management services of this kind are insured services in the United States. There is no evidence that this is the case in British Columbia. Neither Mr. Douglas nor Dr. Chan made inquiries about whether ICBC or a private insurer would pay for this kind of service. Mr. Teed testified that it was difficult to

get ICBC to pay for services to individuals with chronic pain at the existing multidisciplinary Vancouver clinics.

[136] Although Dr. Chan had mentioned the idea of a pain clinic to Lorraine Melvin, a long-time employee, she thought this conversation had taken place two years before the accident. Dr. Chan's accountant, Alan Bruyneel, also recalled a discussion about a pain clinic, (although he was uncertain when this occurred), but he was never asked to prepare any projections for one. No business plan was ever prepared. All of the projections for the pain clinic were prepared for use in the litigation and were not subjected to any independent verification.

[137] The projected revenues and expenses are hypothetical and quite unrealistic. For example, the projected hourly rate for Dr. Chan is twice his highest billing rate. In the first year, Mr. Douglas projected that Dr. Chan would spend five hours a day working in the pain clinic. He could not have done this and also maintained his existing practice. Further, he was already working up to 75 hours a week and while I accept that he had a voracious appetitive for work, Dr. Chan would never have been in a position to devote this kind of time to it.

[138] The assessment of future economic loss is, by its nature, speculative, but the plaintiffs must still establish that there is a real and substantial possibility that Dr. Chan would have established the pain clinic and lost the income that flowed from this. Notwithstanding the lower test for this kind of loss, I find that this has not been proven and I therefore assess these damages as nil.

[139] **Summary of Assessment of Damages**

Non-Pecuniary Damages	$185,000
FLA Claims	$110,000
Special Damages	$153,429
Future Care	$612,544 (15% contingency applied)
Past Income Loss	$343,200

Future Income Loss	$125,754 (15% contingency applied)
Total	**$1,526,808**
Subrogated Claim (agreed)	$3,699.59

[140] The plaintiffs are entitled to pre-judgment interest in accordance with the provisions of the *Courts of Justice Act*. Counsel may arrange a conference call attendance to address the issues of costs, pre-judgment interest or any other matters, including any errors in my calculations.

LAX J.

Released: November 25, 2005

COURT FILE NO.: 95-CU-89403
DATE: 2005/11/25

ONTARIO

SUPERIOR COURT OF JUSTICE

B E T W E E N:

DAVID C. CHAN, BETTY JUNG CHAN, STEPHEN C. CHAN and ANDREW DAVIS CHAN

Plaintiffs

- and -

THE ERIN MILLS TOWN CENTRE CORPORATION, LOBLAWS COMPANIES LIMITED, LOBLAWS INC. and LOBLAWS SUPERMARKETS LIMITED

Defendants

REASONS FOR JUDGMENT

Lax J.

Released: November 25, 2005